

Chapter 6: Symmetries

Quantum Mechanics

1 Introduction to Symmetry

Definition 1.1 (Symmetry). [def:symmetry] A *symmetry* of a system is some transformation, such that the system is invariant before and after the transformation.

Definition 1.2 (Symmetry of Quantum System). [def:symmetry-qm] A *symmetry* of a quantum system is a transformation that leaves the Hamiltonian invariant.

Definition 1.3 (Types of Symmetry). [def:symmetry-types] There are several types of symmetries in quantum mechanics, including:

- *Translation symmetry*: The system is invariant under some spatial translations.
- *Rotation symmetry*: The system is invariant under some rotations.
- *Reflection symmetry*: The system is invariant under reflections on some axis/planes.
- *Inversion symmetry*: The system is invariant under inversion of all coordinates.
- *Time-reversal symmetry*: The system is invariant under the reversal of time.

Theorem 1.1 (Symmetry Condition on the Hamiltonian). [thm:symmetry-condition-hamiltonian] An operator $\hat{U}$ realizes a definition 1.2 of a system with Hamiltonian $\hat{H}$ if and only if $\hat{H}$ is invariant under definition conjugation action:

$$\hat{U}^\dagger \hat{H} \hat{U} = \hat{H}, \quad \text{equivalently (for linear } \hat{U}) \quad [\hat{U}, \hat{H}] = 0.$$

For antiunitary $\hat{U}$ (e.g. time reversal), the second form must be replaced by the appropriate anti-linear condition.

Remark 1.1 (“ $\hat{A}$ Obtained After Applying $\hat{U}$ ”). [rem:conjugation-vocabulary] The phrase “ $\hat{A}$ obtained after applying $\hat{U}$ ” refers to the conjugation $\hat{U}^\dagger \hat{A} \hat{U}$, since this preserves the expectation value.

2 Groups

Definition 2.1 (Group). [def:group] A *group* consists of the following data and assumptions:

- A set G .
- A binary operation $\cdot : G \times G \rightarrow G$ (the codomain being G bakes in closure).
- An identity element $e \in G$.
- An inverse function $(\cdot)^{-1} : G \rightarrow G$, written $g \mapsto g^{-1}$.
- *Associativity*: $(g_1 \cdot g_2) \cdot g_3 = g_1 \cdot (g_2 \cdot g_3)$ for all $g_1, g_2, g_3 \in G$.
- *Identity axiom*: $e \cdot g = g \cdot e = g$ for all $g \in G$.
- *Inverse axiom*: $g \cdot g^{-1} = g^{-1} \cdot g = e$ for all $g \in G$.

A collection with a well-behaved multiplication and inverses.

Example 2.1 (Real Numbers Under Addition). [ex:reals-group] Choose:

- Set $G := \mathbb{R}$ (the real numbers).
- Operation $\cdot := +$ (ordinary addition).
- Identity $e := 0$.
- Inverse $a^{-1} := -a$.

Associativity, identity, and inverse axioms follow from the usual properties of addition.

Example 2.2 (Boolean Set Under XOR). [ex:boolean-group] Choose:

- Set $G := \{\text{tt}, \text{ff}\}$ (equivalently $\{0, 1\}$).

- Operation $\cdot := \oplus$ (XOR, equivalent to addition mod 2).
- Identity $e := \text{ff}$ (equivalently 0).
- Inverse $g^{-1} := g$ (each element is self-inverse).

All axioms hold by direct verification.

Definition 2.2 (Continuous Group). [def:continuous-group] A group is *continuous* (a *Lie group*) if its elements are parametrized by a [continuous parameter space](#), and multiplication and inversion vary smoothly with the parameters. The real numbers under addition is an example.

Definition 2.3 (Discrete Group). [def:discrete-group] A group is *discrete* if its elements form a [discrete set](#) (finite or [countable](#)), with no continuous parameter connecting distinct elements. The Boolean set under XOR is an example.

Definition 2.4 (Continuous Symmetry). [def:continuous-symmetry]

Definition 2.5 (Discrete Symmetry). [def:discrete-symmetry]

3 Translation Operator

Definition 3.1 (Translation Operator). [def:translation-operator] Intuitively, a *translation operator* $\hat{T}(\vec{r}_0)$ (translates a state by a vector $\vec{r}_0$) acts on a state $|\psi\rangle$ should let $|\psi\rangle$ be translated by a vector $\vec{r}$, i.e.

$$\hat{T}(\vec{r}_0) |\psi\rangle = |\psi(\vec{r} - \vec{r}_0)\rangle$$

Definition 3.2 (Translation Operator (1D)). [def:translation-operator-1d] In one dimension, the translation operator $\hat{T}(a)$ shifts a state by a real distance $a \in \mathbb{R}$. Its action on a position-space wavefunction is

$$\hat{T}(a) \psi(x) = \psi(x - a).$$

Theorem 3.1 (Translation Operators Form a Symmetry Group). [thm:translation-symmetry-group] Suppose $\hat{T}(a)$ is a symmetry operator of a Hamiltonian $\hat{H}$ for every $a \in \mathbb{R}$. Then the collection of translation operators forms a group under operator composition:

- Set $G := \{\hat{T}(a) : a \in \mathbb{R}\}$.
- Operation $\cdot$: operator composition, $\hat{T}(a) \cdot \hat{T}(b) = \hat{T}(a + b)$.
- Identity $e := \hat{T}(0) = \hat{I}$.
- Inverse $\hat{T}(a)^{-1} := \hat{T}(-a)$.

All axioms follow from addition on $\mathbb{R}$.

Proof.

□

Theorem 3.2 (Exponential Form of the Translation Operator). [thm:translation-exponential] The one-dimensional translation operator can be written as:

$$\hat{T}(a) = e^{-ia\hat{p}/\hbar}$$

Proof. Expand the exponential in a Taylor series at 0

$$\hat{T}(a) = e^{-ia\hat{p}/\hbar} = \sum_{n=0}^{\infty} \frac{(-ia/\hbar)^n}{n!} \cdot \hat{p}^n = \sum_{n=0}^{\infty} \frac{(-ia\hat{p})^n}{n!\hbar^n}$$

According to the definition of the momentum operator, $\hat{p} = -i\hbar \frac{d}{dx}$, so we can substitute this into the series expansion:

$$\hat{T}(a) = \sum_{n=0}^{\infty} \frac{(-ia(-i\hbar \frac{d}{dx}))^n}{n!\hbar^n} = \sum_{n=0}^{\infty} \frac{(-a(\frac{d}{dx}))^n}{n!}$$

So,

$$\hat{T}(a)\psi(x) = \sum_{n=0}^{\infty} \frac{(-a(\frac{d}{dx}))^n}{n!} \psi(x) = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}(x)$$

Then, we do a Taylor expansion of $\psi(x)$ centered at $x + a$:

$$\psi(x) = \sum_{n=0}^{\infty} \frac{\psi^{(n)}(x+a)}{n!} (x - (x+a))^n = \sum_{n=0}^{\infty} \frac{\psi^{(n)}(x+a)}{n!} (-a)^n = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}(x+a)$$

Substituting $x \rightarrow x - a$ in the previous identity,

$$\psi(x-a) = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}((x-a)+a) = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}(x).$$

Which means,

$$\hat{T}(a)\psi(x) = \psi(x-a)$$

□

Theorem 3.3 ($\hat{T}(a)$ is a Unitary Operator). [thm:translation-unitary] *The translation operator $\hat{T}(a)$ is a unitary operator, i.e. $\hat{T}^\dagger(a) = \hat{T}^{-1}(a) = \hat{T}(-a)$.*

Remark 3.1 (Intuition about). [rem:intuition-about]

Proof. **TODO:** Proof to be written after clarifying preferred approach with the lecturer — direct from the action definition (definition 3.2), or via the exponential form (theorem 3.2). Email sent September 23, 2026. □

Theorem 3.4 (Position Operator After Applying Translation). [thm:position-under-translation] *Applying $\hat{T}(a)$ to the position operator $\hat{x}$ (see remark 1.1) gives*

$$\hat{T}(a)^\dagger \hat{x} \hat{T}(a) = \hat{x} + a\hat{I}.$$

Proof. Define $\hat{x}' := \hat{T}(a)^\dagger \hat{x} \hat{T}(a)$, and consider its action on a wavefunction $\psi(x)$:

$$\hat{x}'\psi(x) = \hat{T}(a)^\dagger \hat{x} \hat{T}(a) \psi(x)$$

Since operators are associative, we have

$$\hat{x}'\psi(x) = \hat{T}(a)^\dagger \hat{x} (\hat{T}(a) \psi(x)) = \hat{T}(a)^\dagger (\hat{x} \psi(x-a)) = \hat{T}(a)^\dagger (x \psi(x-a))$$

Treat $g(x) = \hat{x} \psi(x-a)$, so, $\hat{T}(a) g(x) = g(x) = x \hat{T}(a) \psi((x-a)-a)$.

Notice that $\hat{T}(a)$ is a unitary operator, ie $\hat{T}(a)^\dagger = \hat{T}(a)^{-1} = \hat{T}(-a)$, we have

$$\hat{x}'\psi(x) = \hat{T}(-a)^\dagger (x \psi(x-a)) = (x+a) \psi((x+a)-a) = (x+a) \psi(x)$$

Therefore, $\hat{x}' = \hat{x} + a\hat{I}$. □

Theorem 3.5 (Momentum Operator After Applying Translation). [thm:momentum-under-translation] *Applying $\hat{T}(a)$ to the momentum operator $\hat{p}$ (see remark 1.1) gives*

$$\hat{T}(a)^\dagger \hat{p} \hat{T}(a) = \hat{p}.$$

Proof. □

4 Rotational Symmetry

5 Parity/Inverse Symmetry

Example 5.1 (Parity Group). [ex:parity-group] Choose:

- Set $G := \{\hat{I}, \hat{P}\}$.
- Operation $\cdot$: operator composition.
- Identity $e := \hat{I}$.
- Inverse $\hat{P}^{-1} := \hat{P}$ (since $\hat{P}^2 = \hat{I}$, parity is self-inverse).

A discrete group with two elements.

6 Time-Reversal Symmetry